

Properties at a glance

Three properties belong to a vector on its own; three exist only between a pair. The split matters because the zero vector behaves differently on either side of it.

UNCONDITIONAL

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i **Magnitude** $\|\mathbf{v}\| =$

UNCONDITIONAL

REQUIRES intrinsic — belongs to one vector

A single non-negative number: the length of the arrow, independent of where it is drawn. It is zero exactly for the zero vector, which is the property positive definiteness of the **dot product** records.

$$\sqrt{v_1^2 + v_2^2 + \dots + v_n^2}$$

iii **Dimension** $\mathbf{v} \in \mathbb{R}^n \Rightarrow \dim = n$

UNCONDITIONAL

REQUIRES intrinsic — fixed by the space

The component count, decided by the space rather than the vector. No operation on vectors changes it — addition and scaling stay inside $\mathbb{R}^n$, which is what makes it a **vector space** rather than merely a set.

CONDITIONAL

n

4

ii **Direction** $\hat{\mathbf{v}} =$

CONDITIONAL

HOLDS WHEN $\mathbf{v} \neq \mathbf{0} \Rightarrow \hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|}$

FAILS WHEN $\mathbf{v} = \mathbf{0}$ — division by zero

The unit vector pointing the same way. The zero vector has no direction at all, and this is not a convention that could have gone otherwise: the definition divides by a magnitude of zero. Every later statement about $\mathbf{0}$ traces back here.

iv **Equality** $\mathbf{a} = \mathbf{b} \Leftrightarrow a_i = b_i$ for every i

CONDITIONAL

HOLDS WHEN both in the same $\mathbb{R}^n$

All-or-nothing: one differing component breaks it. Vectors in different spaces are not unequal so much as incomparable — the question does not arise.

v **Parallelism** $\mathbf{a} \parallel \mathbf{b} \Leftrightarrow \mathbf{a} = c\mathbf{b}$ for some scalar c

CONDITIONAL

HOLDS WHEN either vector may be $\mathbf{0}$

COMMON ERROR assuming parallel means same direction — $c < 0$ gives opposite directions

A relation between two vectors, not a property of either. The zero vector is parallel to everything by convention, which is consistent because $\mathbf{0} = 0 \cdot \mathbf{b}$ holds for every $\mathbf{b}$. In $\mathbb{R}^n$ the **cross product** detects this: parallel vectors have $\mathbf{a} \times \mathbf{b} = \mathbf{0}$.

vi **Orthogonality** $\mathbf{a} \perp \mathbf{b} \Leftrightarrow \mathbf{a} \cdot \mathbf{b} = 0$

CONDITIONAL

HOLDS WHEN either vector may be $\mathbf{0}$

Also relational. The zero vector is orthogonal to everything, since $\mathbf{0} \cdot \mathbf{b} = 0$ always — so it is simultaneously parallel and orthogonal to every vector, the only one for which both hold. See **orthogonality** for what this makes possible.

THE ZERO VECTOR CASE

$\mathbf{0} \cdot \mathbf{b} = 0$ for every $\mathbf{b} \rightarrow \mathbf{0} \perp \mathbf{b}$

$\mathbf{0} = 0 \cdot \mathbf{b}$ for every $\mathbf{b} \rightarrow \mathbf{0} \parallel \mathbf{b}$

Notice how often the zero vector is the exception. It has no direction, is parallel to everything, and is orthogonal to everything — three conventions that look arbitrary until you see they all follow from $\mathbf{0}$ having magnitude zero and no defined unit vector.